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Multi-Agent Deep Reinforcement Learning Based Consistency Control in Train Platoon Systems

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***Abstract*—The train platoon (TP) in the urban rail transit (URT) system has been regarded as a promising solution to improve the transportation capacity between metro stations. Moreover, TP has a great potential to effectively adapt to the high dynamic flow of passengers during major events or festivals. Unfortunately, the promotion of TP is challenged due to the auto-block system in the traditional URT system as well as the complicated and dynamic operation environment. This paper investigates a cooperative control method to realize the consistency of TP in the premise of anti-interference. The multi-agent system of TP based on leader-follower typology is designed to achieve the operation goals in terms of distance and velocity. A novel distributed multi-agent deep reinforcement learning algorithm is proposed to obtain the optimal control strategy, where the artificial potential field is structured to provide prior knowledge of the environment. Compared with the conventional Q-learning based methods, simulation results validate that the proposed algorithm not only benefits the achievement of the operation goal, but also prevents the unfavorable interference from the external system.**

***Index Terms*—Multi-agent deep reinforcement learning, train platoons, artificial potential field, cooperative control.**

I. Introduction

Communication-based train control system (CBTC) serves as the core of urban rail transit (URT) system nowadays. Under CBTC mechanism, two transportation problems urge to be solved. Firstly, with the increasing pressure of overwhelming passenger flow, the capacity of railway transportation can not satisfy the quantity of passengers virtually. Overload of trains happens occasionally, which may threaten life security of passengers. Secondly, regarded as the phenomenon of tidal passenger flows, the direction into the city requires more trains than the counterpart in the morning peak. Instead, both directions are allocated with identical numbers of train sets, in which case demand exceeds supply in the direction of city-entering and vice versa, causing transportation resource waste.

Based on aforementioned problems, a novel train platoon (TP) system is structured to increase the transportation capacity and accommodate the high dynamic flow of passengers. In order to shorten the distance between adjacent trains in TP, train-to-train is introduced as another communication method in addition to train-to-infrastructure to reduce the communication delay [1]. However, the auto-block system of the control center in CBTC restrains the relative distance of trains [2]. Therefore, auto-control based on commands from the control center is possible to be substituted for smart control based on artificial intelligence. To facilitate the matching between trains and passengers, trains in TP utilize wireless communication as a virtual coupler replacing the mechanical coupler so that TP can enroll or deprive trains according to the quantity of passengers flexibly [3]. Nevertheless, the dynamic motion of trains and the complicated environment pose a threat to operation security of TP. In this paper, we consider operation security is achieved if TP arrives at the destination successfully without out-of-range accidents and break-down caused by unpredictable interference.

Considering that TP serves as a multi-agent system (MAS), deep reinforcement learning (DRL) is regarded as an appropriate AI control method of TP in recent studies. Most existing works adopt centralized DRL based on global information on considerable states [4], whose dimension increases intensely when participants enroll in TP, leading to unforeseen computation complexity. Meanwhile, DRL lacks the prior knowledge resulting in poor training efficiency. Motivated by the aforementioned challenges of DRL, we propose a distributed multi-agent deep reinforcement learning (multi-agent DRL) to control agents locally and increase the training speed by the pre-designed reward function. Compared with general DRL, the proposed multi-agent DRL has greater advantage of saving computation resource and improving training efficiency, which is more suitable to the TP system with multiple agents.

A. Related Work

Researchers have developed various control methods to operate vehicle platoons. In early studies, researchers focused on the motion of platoons, including formulation of platoon and path planning [5]. Recently, robust performance of vehicle operation has been considered based on cybernetics control method. Considering possible interior error or exterior interference, control methods were developed by [6]–[8] to realize error detection and correction. Specifically, Wang *et al.* proposed a distributed model predictive control algorithm to tackle time-varying constraints derived from abnormal communication by switching communication topology and control topology [6]. Alotibi *et al.* considered the critical threats of false information from the platoon leader, and proposed a real-time anomaly detection mechanism that combined statistical

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learning and kinematics laws [7]. Meanwhile, information sharing method was designed in cooperative adaptive cruise control (CACC) model to support the detection and improve the reliability. Considering cyber-security of communication networks between vehicles, a distributed security control approach driven by data was proposed to prevent denial-of-service attacks, which was suitable for nun-model system [8].

Meanwhile, RL-based control method has substituted for conventional cybernetics increasingly. Robust performance of vehicle operation was improved by [9]–[12]. Specifically, Li $et\ al.$ adopted a curriculum learning approach to reduce crashes during training exploration [9]. Sui $et\ al.$ proposed two-stage training framework based on DRL to balance the comprehensive problem of formation maintenance and collision avoidance [10]. Zhang $et\ al.$ integrate soft actor-critic algorithm with Lyapunov function to guarantee stability of motion planning of autonomous vehicles [11]. Peake $et\ al.$ developed a novel CCAC method based on multi-agent DRL to optimize decision-making of communication and improve the string stability of driving in platoons [12]. Moreover, [9], [13] contributed to save resource of platoon system. Concretely, Meng $et\ al.$ designed a communication proximal policy optimization to tackle redundant information derived from multi-agent RL algorithms [9]. Du $et\ al.$ employed DRL for energy management of the series electric vehicle and created heuristic method for efficient sampling [13]. Furthermore, adaptive control methods were developed for more complex scenarios. With regard to motion control under the roundabout, Zhang $et\ al.$ developed an optimization embedded RL to achieve adaptive decision-making [14]. Taking into account the platoon containing both autonomous vehicles and human-driven vehicles, an adaptive optimal control method for the former was proposed to tackle heterogeneous driver behavior, which adapted to different platoon dynamics [15].

B. Contributions

Due to the dynamic motion of multiple trains in the complicated environment, achieving an efficient cooperative control of TP is a challenging problem. In this paper, we propose a distributed reinforcement learning method to solve the consistency control problem with security guarantee. The main contributions of this paper are listed as follows:

- We formulate a multi-objective problem in the TP system based on a leader-follower typology, where an artificial potential field (APF) method is adopted to unify a variety of force. Unlike APF in previous studies, we design twin potential fields in terms of distance and velocity difference between adjacent trains separately, and the value of potential field force is limited to suit for reality.
- By designing reward function based on APF to provide prior knowledge and improve training efficiency, we develop a distributed multi-agent DQN algorithm which allows agents to prevent overwhelming state information. Different from the past studies, the reward function is built according to characteristics of different states, which accelerates the training process to a large degree.

TABLE I: Summary of Symbols and Main Notations

$\mathcal{N}$, N	Set of trains in TP, and size of the set
n, x_n	Operation distance from beginning of train n
$x_{n-1,n}$	Relative distance of train $n-1$ and train n
$e_{n-1,n}$	Distance error of train $n-1$ and train n
d_{goal}	Goal distance distance between adjacent trains
$d_{\max}$	Maximum distance between adjacent trains
v_n	Velocity of the n-th train
$v_{n-1,n}$	Velocity difference of train $n-1$ and train n
$v_{n,\max}$	Maximum velocity of the n-th train
$u_{nx}(e_{n-1,n})$	Position controller of the n-th train
$u_{nv}(v_{n-1,n})$	Velocity controller of the n-th train
k	Period number of wireless communication
$\mathcal{S}$, $\mathcal{A}$	Set of operation states, and set of actions
s_n	State vector of the n-th train
$\delta_{n-1,n}$, $\nu_{n-1,n}$	Distance state and velocity state
a_n	Acceleration and action of the n-th train
r_n	Reward function of the n-th train
$\mathbf{w}_n$, $\mathbf{w}_n^-$	Weights of Q-network and Q-target network
$\mathcal{I}$	Sample set selected for every training
I	Size of the sample set
ϵ	Probability to select a random action
γ	Discount factor to adjust the sensitivity
$Q^*(s_n, a_n)$	Q value function
$Q(s_n, a_n; \mathbf{w}_n)$	Estimated Q value function
$L_{ni}(\mathbf{w}_{ni})$	Mean-square error method based loss function
y_n	Target function in Q-network

- We further convert communication typology from uni-direction to bi-direction to constrain distance between adjacent trains so as to reduce out-of-range communication. A virtual leader is employed to regulate the prescriptive schedule and route, and the reward is upgraded to be multi-standard to achieve the operation goal.
- Compared with Q-learning, simulation results show that our proposed learning algorithm can achieve better performance in terms of consistency control, decoupling reduction and interference suppression. Meanwhile, bi-direction communication method makes a difference to out-of-range communication precaution remarkably.

The rest of this paper is organized as follows. Section II describes the structure of the TP system and the formulated multi-objective problem. Section III elaborates on the proposed distributed multi-agent DQN algorithm. Section IV presents numerical results of the proposed algorithm compared with Q-learning for performance evaluation. Finally, section V provides conclusion to this paper.

II. System Model and Problem Formulation

In this paper, we focus on the entire journey of TP from one destination to another. In the premise of the successful arrival at destination, the operation goal of TP is to achieve the consistence of velocity and distance between adjacent trains simultaneously, which is defined as the cooperative consistency control of TP. The operation goal of TP is able to be converted into the operation goal of followers, which is to achieve zero velocity difference and the predefined goal distance simultaneously between every follower and the predecessor. Note that the predefined goal distance is identical in terms of every follower.

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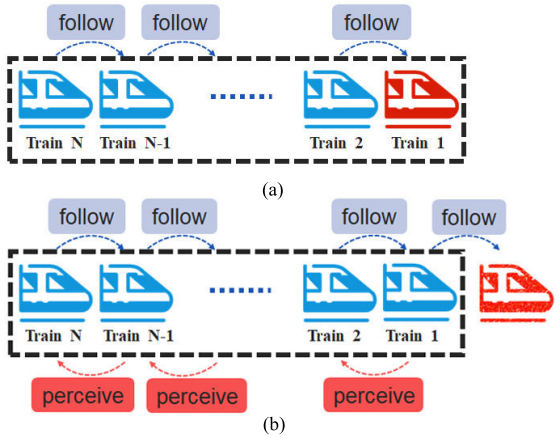


Fig. 1: System model of TP with N trains. (a) Model of uni-direction communication. (b) Model of bi-direction communication.

A. System Model

As illustrated in Fig. 1(a), the leader-follower typology is applied to the formation of TP. N denotes the size of TP indexed by $\mathcal{N} = \{1, 2, ..., N\}$. Members of TP are divided into leaders and followers. Leaders play a role of leading direction, including trains in TP except the N-th train. Followers are the trains behind leaders, including trains in TP except the first train. To be simplified, any follower mentioned below is denoted as the n-th train, and $\forall n \in \mathcal{N}\backslash\{1\}$ if not instructed particularly. With the exception of the first and the N-th train, every train in TP is regarded as not only a leader but also a follower, which follows the predecessor and leads the successor. Every follower communicates with the predecessor leader solely. Followers depend on the operation states of their leaders to decide the control strategies.

We model the complex dynamics environment of TP as an artificial potential field (APF) where potential field force is the only variety of force that trains take [16]. Two aspects are considered to design the APF. Firstly, the zero potential point is required to be the operation goal of TP. Considering the local optimization of APF, the potential field function requires a sole minimum. Secondly, the potential field force function is required to reach zero at the zero potential point. In other words, trains maintain the goal operation state. In other situations, the potential field force of a train points to the zero potential point, which means the potential field force can be attractive or repulsive. The potential field force drives TP to achieve the operation goal. Considering practical situation, the potential field force function is required to be bounded. Based on the aforementioned aspects, $\ln(\cosh(\cdot))$ is selected as the potential field function and $\nabla \ln(\cosh(\cdot)) = \tanh(\cdot)$ serves as the potential field force function [1]. In view of the operation goal, twin potential fields are designed and constitute the entire artificial environment.

We define x_n as the operation distance of the n-th train, and the distance between adjacent trains is $x_{n-1,n} = x_{n-1} - x_n$. d_{goal} is defined as the goal distance between adjacent trains, and the distance error between them is $e_{n-1,n} = x_{n-1,n} - d_{\text{goal}}$. v_n is defined as the velocity of the n-th train, and the velocity difference between adjacent trains is $v_{n-1,n} = v_{n-1} - v_n$. The potential field force function is considered as a controller of the n-th train, which is described as

$$u_n{}^{APF} = \underbrace{\alpha_1 \sum_{n'=1}^{N} a_{nn'} \tanh(e_{nn'})}_{u_{nx}(e_{nn'})} + \underbrace{\alpha_2 \sum_{n=1}^{N} a_{nn'} \tanh(v_{nn'})}_{u_{nv}(v_{nn'})}, \tag{1}$$

where $\forall n', n \in \mathcal{N}, n' \neq n$, $u_{nx}(e_{n,n'})$ is a position controller, which controls the n-th train to reach d_{goal}, $u_{nv}(v_{nn'})$ is a velocity controller, which controls the n-th train to achieve the consistent velocity with the predecessor, $u_n{}^{APF}$ is the compound controller consisting of the two, α_1 and α_2 are positive parameters which adjust the amplitudes of the controllers, $a_{nn'}$ is an element in the matrix $A_m = [a_{nn'}]_{N\times N}$ of the formation typology of TP. In this model, $A_m = \begin{bmatrix} 0 & A_{N-1,N-1} \\ 0 & 0 \end{bmatrix}$, where $A_{N-1,N-1}$ is an adjacent matrix and hence the controller of followers is simplified as

$$u_n{}^{APF} = \underbrace{\alpha_1 \tanh(e_{n-1,n})}_{u_{nx}(e_{n-1,n})} + \underbrace{\alpha_2 \tanh(v_{n-1,n})}_{u_{nv}(v_{n-1,n})}, \tag{2}$$

where $\forall n \in \mathcal{N}\backslash\{1\}$. The controller of the n-th train in TP is described as

$$u_n = \begin{cases} u_n^{ATO}, & n = 1 \\ u_n^{APF}, & n \neq 1 \end{cases} \tag{3}$$

where u_1^{ATO} denotes the controller of the first train in the automatic train operation (ATO) module.

In this paper, the operation journey of the first train controlled by ATO is divided into three stages: traction, cruising and braking, and the operation stage of the n-th train is defined by the real-time stage of the first train. At the traction stage, v_1 increases from 0 to the maximum $v_{1,\max}$, and $x_1^A = v_{1,\max}^2 / 2a_1^A$. At the cruising stage, v_1 remains $v_{1,\max}$, and $x_1^B = v_{1,\max} t_1^B$. At the braking stage, v_1 decreases from $v_{1,\max}$ to 0, and $x_1^C = -v_{1,\max}^2 / 2a_1^C$, where a_1 denotes the acceleration of the first train, A, B and C denote the three stages successively.

B. Problem Formulation

The operation goal of TP is to achieve the cooperative consistency control in the premise of the constraints for security guarantee. The operation goal and the constraints are formulated as

$$\text{goal}: \begin{cases} \lim\limits_{k\to\infty} x_{n-1,n}^k(a_n^k) = d_{\text{goal}}, \\ \lim\limits_{k\to\infty} v_{n-1,n}^k(a_n^k) = 0, \end{cases} \tag{4}$$

$$\text{constraints}: \begin{cases} 0 \leq x_{n-1,n}^k \leq d_{\max}, \forall n, \forall k, \\ 0 \leq v_n^k \leq v_{n,\max}, \forall n, \forall k, \\ a_{\text{maxb}} \leq a_n^k \leq a_{\text{maxt}}, \forall n, \forall k, \end{cases} \tag{5}$$

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where k denotes the period number of wireless communication between adjacent trains as a positive integer, the variables with superscript k refer to the variables in the k-th communication period, $d_{\max} = \max(x_{n-1,n})$ which is determined by the maximum range of wireless communication, $v_{n,\max} = \max(v_n)$ which is not identical for every train to obtain better performance, $a_{\max b}$ and $a_{\max t}$ denote the maximum deceleration and the maximum acceleration of the n-th train, which are predefined and identical for every train.

However, It's difficult to obtain an optimal control strategy due to the following reasons. Firstly, predecessors are part of the operation environment of TP resulting in the variability and complexity of surroundings. Secondly, the overwhelming data derived from TP brings excessive pressure on computation and storage of the transportation system. Thirdly, traditional auto-control method inclines to be stuck in local optimization. Therefore, it's crucial to introduce reinforcement learning in the cooperative control of TP.

III. Multi-agent Deep Q-Network for Cooperative Consistency Control

To address the above problems, we turn to developing DRL-based control method for followers, which is capable to search the global optimal strategy. However, due to zero prior knowledge of environment and random action-choosing at the start of training, DRL is of low training efficiency. Moreover, to deal with the MAS, the general DRL produces overwhelming data leading to excessively complex computation. To solve these problems, we develop a distributed multi-agent DRL algorithm. Distributed DRL means every agent in TP is trained separately and has personal states, actions and rewards differently, instead of being trained jointly and sharing identical states and rewards. Compared with general DRL, the reward function of multi-agent DRL is delicately designed in accordance with APF, by means of which APF plays a role of prior knowledge provider. Deep Q-Network (DQN) is adopted due to the significant competence in fitting nonlinear function and saving storage space.

A. MDP Formulation

Multi-agent DQN agents in TP interact with the environment and transit to the next operation state, which can be modeled as a Markov decision process (MDP). In this section, we formulate the state space $\mathcal{S}$, the action space $\mathcal{A}$ and the reward function of the n-th train r_n. Note that we assume the state transition probability is 1.

State space: We assume $\mathcal{S}$ denotes the state space of the n-th train. The state of the n-th train denoted as s_n is a 2-dimension vector consisting of $x_{n-1,n}$ and $v_{n-1,n}$. Due to the continuity of $x_{n-1,n}$ and $v_{n-1,n}$, discretization is of great importance. In this paper, the value range of $x_{n-1,n}$ is divided equally into X segments denoted as $\delta_{n-1,n} \in \{1, 2, ..., X\}$. Similarly, the value range of $v_{n-1,n}$ is divided equally into V segments denoted as $\nu_{n-1,n} \in \{1, 2, ..., V\}$. Hence $s_n = (\delta_{n-1,n}, \nu_{n-1,n}), \forall s_n \in \mathcal{S}$, and the size of state space is XV.

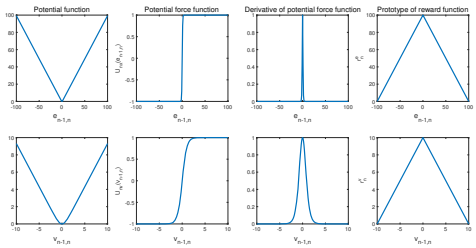


Fig. 2: Prototypes of twin reward functions r_n^e and r_n^v.

Action space: We assume $\mathcal{A}$ denotes the action space of the n-th train. We regard the acceleration of the n-th train as an action denoted as a_n, $\forall a_n \in \mathcal{A}$. Due to the continuity of accelerations and simplicity of the problem, we assume the n-th train is equipped with three gears consisting of acceleration gear, idle gear and deceleration gear, which are set as $\mathcal{A} = \{a^+, 0, a^-\}$, and the size of action space is 3.

Reward function: Reward function is designed in accordance with aforementioned controllers derived from APF. We assume the reward function of the n-th train is $r_n = f(e_{n-1,n}, v_{n-1,n})$ consisting of $r_n^e = f_e(e_{n-1,n})$ and $r_n^v = f_v(v_{n-1,n})$. As a core discipline of reward-designing, when a follower is closer to the operation goal, the reward value is larger. In other words, when a follower is closer to the zero potential point or takes less potential field force, the reward value is larger. As shown in Fig. 2, we simplify the derivative curve of potential field forces function as prototypes of twin reward functions. Based on the prototypes, specific disciplines to design r_n are as follows:

Fixed reward value: For the sake of improving the efficiency of training, some fixed and larger reward values are designed for the operation goal achievement and security problems separately to select a better action. The operation goal achievement is equivalent to $e_{n-1,n} = 0$ or $v_{n-1,n} = 0$, corresponding to $r_n^e = R$ or $r_n^v = R$, where R is a fixed positive large value. Security problems contain collision, out-of-range and reverse, which are separately denoted as $e_{n-1,n} < -d_{\text{goal}}$, $e_{n-1,n} > d_{\max} - d_{\text{goal}}$ corresponding to $r_n^e = -R$, and $v_n < 0$ corresponding to $r_n^v = -R$. When a security problem emerges, training failure happens.

Safe states and edge states: The edge state is defined as a state where the training is close to failure, while the safe state is the one except edge states and training failure states in $\mathcal{S}$. To facilitate the training process, we design the safe state corresponds to positive reward while the edge state corresponds to negative reward. Specifically, edge states of $e_{n-1,n}$ include $-d_{\text{goal}} \le e_{n-1,n} < 0$ and $d_{\max} - d_{\text{goal}} - D < e_{n-1,n} \le d_{\max} - d_{\text{goal}}$, where D is a fixed value to define the range of edge state of $e_{n-1,n}$. To achieve the optimal strategy quickly during transaction and braking stages particularly, we define edge states of $v_{n-1,n}$ by introducing $w_{\max 1}$ and $-w_{\max 2}$ as shown in Fig. 3.

For transaction stage with $v_{n-1,n} \ge 0$ constantly, $w_{\max 1}$

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denotes the maximum of $v_{n-1,n}$ if no out-of-range happens. More specifically, from the moment of $v_{n-1,n} = w_{\max 1}$, if the agent accelerates continually until $v_{n-1} = v_{n-1,\max}$ with $v_{n-1,n} = 0$, the edge state $e_{n-1,n} = d_{\max} - d_{\text{goal}}$ is obtained exactly. In other words, $v_{n-1,n} > w_{\max 1}$ definitely leads to out-of-range. Therefore, $w_{\max 1}$ is formulated as $w_{\max 1} = v_{n-1} - v_n$, where v_{n-1} and v_n satisfy $x_{n-1,n} + 0.5((v_{n-1,\max} - v_n)t_n - (v_{n-1,\max} - v_{n-1})t_{n-1}) = d_{\max}$.

For braking stage with $v_{n-1,n} \leq 0$ constantly, $-w_{\max 2}$ denotes the minimum of $v_{n-1,n}$ in the premise of no collision. More specifically, from the moment of $v_{n-1,n} = -w_{\max 2}$, if the agent selects deceleration action continually until $v_{n-1} = v_n = 0$, the edge state $e_{n-1,n} = -d_{\text{goal}}$ is obtained exactly. In other words, $v_{n-1,n} < -w_{\max 2}$ definitely leads to collision. Therefore, $-w_{\max 2}$ is formulated as $-w_{\max 2} = v_{n-1} - v_n$, where v_{n-1} and v_n satisfy $x_{n-1,n} - 0.5(v_{n-1}t_{n-1} - v_n t_n) = 0$. Note that $v_{n-1,n} > w_{\max 1}$ and $v_{n-1,n} < -w_{\max 2}$ do not lead to training failure immediately, which are reasonable to be considered as edge states. It is worth mentioning that $w_{\max 1}$ and $-w_{\max 2}$ are time-varying, which is not a problem for identification of edge states.

Combination of r_n^e and r_n^v: To intergrate r_n^e and r_n^v to r_n reasonably, we define the priority of the negative one is higher than that of the positive one. More specifically, as long as anyone of the two is negative, the positive one is hidden.

In accordance with three principles above, the reward function of the n-th train in TP is formulated as

$$r_n = \begin{cases} r_n^e + r_n^v, & r_n^e r_n^v > 0 \\ r_n^e, & r_n^e < 0, r_n^v > 0 \\ r_n^v, & r_n^v < 0, r_n^e > 0 \end{cases} \tag{6}$$

Note that if r_n^e or r_n^v equals to zero, r_n is consistent with the non-zero one. To be simplified, we define $d_{\text{m}} = d_{\max} - d_{\text{goal}}$. r_n^e and r_n^v are separately formulated as

$$r_n^e = \begin{cases} d_{\text{m}} d_{\text{goal}}^{-1} e_{n-1,n}, & -d_{\text{goal}} \leq e_{n-1,n} < 0 \\ -e_{n-1,n} + d_{\text{m}}, & 0 < e_{n-1,n} \leq d_{\text{m}} - D \\ d_{\text{m}}((d_{\text{m}} - 1)D^{-1} - 1), & d_{\text{m}} - D < e_{n-1,n} \leq d_{\text{m}} \\ R, & e_{n-1,n} = 0 \\ -R, & \text{else} \end{cases} \tag{7}$$

$$r_n^v = \begin{cases} 2v_{n-1,n}, & v_{n-1,n} < -w_{\max 2} \\ v_{n-1,n} + v_{n,\max}, & -w_{\max 2} \leq v_{n-1,n} < 0 \\ -v_{n-1,n} + v_{n,\max}, & 0 < v_{n-1,n} \leq w_{\max 1} \\ -2v_{n-1,n}, & v_{n-1,n} > w_{\max 1} \\ R, & v_{n-1,n} = 0 \\ -R, & v_n < 0 \end{cases} \tag{8}$$

B. Multi-Agent Deep Q-Network

As illustrated in Fig.4, every agent possesses a personal Q-network trained respectively. Q-network is a deep neural network with three fully connected layers. At the k-th step in a training, the n-th train observes current state s_n^k viewed as a 2-dimension input to Q-network. The second layer of

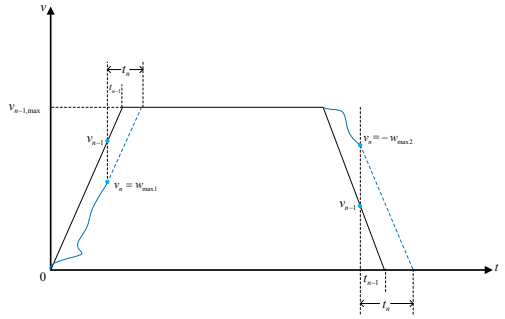


Fig. 3: The simplified illustration of $w_{\max 1}$ and $-w_{\max 2}$.

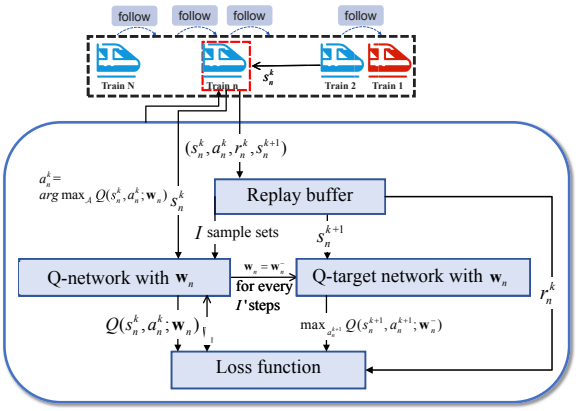


Fig. 4: Framework of the proposed DQN algorithm.

Q-network is a hidden layer, the weight $\mathbf{w}_n$ of which is initialized randomly. Q-network approximates the estimated Q value function by $Q^*(s_n, a_n) \approx Q(s_n, a_n; \mathbf{w}_n)$, where

$$Q(s_n, a_n; \mathbf{w}_n) = \mathbb{E}\left[\sum_{\kappa=k}^{\infty} = \gamma^{\kappa-k} r_n^{\kappa} \mid s_n^k, a_n^k; \mathbf{w}_n\right]. \tag{9}$$

By means of the computation of the hidden layer, the approximate Q-values of three actions in $\mathcal{A}$ are obtained simultaneously as a 3-dimension output. The agent selects the action a_n^k possessing the maximum Q-value with the probability of $1 - \epsilon$, while it selects the action in $\mathcal{A}$ randomly with ϵ. The immediate corresponding reward r_n^k is obtained in accordance with the aforementioned reward function. Simultaneously, the state of the agent transits to the next state s_n^{k+1}.

Experience replay method is employed to distort the correlation of samples and improve the utility efficiency of previous experience. The sample set derived from the k-th step $(s_n^k, a_n^k, r_n^k, s_n^{k+1})$ is preserved in a relay buffer with the capability of B samples. In accordance with the first in first out (FIFO) principle, if the replay buffer is filled and the new sample enrolls, the oldest sample set is expelled from the buffer. We select I samples from the replay buffer for every training indexed by $\mathcal{I} = \{1, 2, ..., I\}$.

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Considering the non-stability of $Q^*(s_n, a_n)$, Q-target network is employed to approximate the actual Q value function known as the target y_n with the fixed weight $\mathbf{w}_n^-$. The loss function based on mean-square error method is formulated as

$$L_{ni}(\mathbf{w}_{ni}) = \mathbb{E}[(y_{ni}^k - Q(s_{ni}^k, a_{ni}^k; \mathbf{w}_n))^2], \forall i \in \mathcal{I}, \forall n, \quad (10)$$

the target y_{ni}^k is formulated as

$$y_{ni}^k = \begin{cases} r_{ni}^k + \gamma \max_{a_{ni}^{k+1}} Q(s_{ni}^{k+1}, a_{ni}^{k+1}; \mathbf{w}_n^-), & k \neq k_\mathrm{t} \\ r_{ni}^k, & k = k_\mathrm{t} \end{cases} \quad (11)$$

where $\gamma \in (0, 1]$ denotes the discount factor to adjust the sensitivity to predicted reward, k_t denotes the terminal period of communication. To minimize the loss function, the weights $\mathbf{w}_n$ is updated by gradient descent method. Initially, sample sets in the replay buffer is fewer than I, training happens at every step employing every sample set, otherwise training process is repeated intermittently with the interval of I steps. Q-target network is identical to Q-network initially, the weight $\mathbf{w}_n^-$ of which is replaced by $\mathbf{w}_n$ with the interval of I' steps.

The agent interacts with the environment and produces one sample set. The process is repeated per step. Hundreds of steps make up one training episode. An entire training episode begins when the agent starts to operate, and ends up with training failure or training success defined as successful arrival at the destination with cessation of every agent. If the current episode fails, then all the variables are initialized and train the current episode a second time until it succeeds. If the current episode succeeds, the training moves on to a new episode. Note that from the initial value ϵ_0, ϵ decreases by ϵ_p per successful episode until obtain the minimum value ϵ_min. In this way, exploration at the beginning and anti-interference ability are achieved, and meanwhile negative influence on efficiency of training is avoided.

C. Extension to Bi-direction Communication

As illustrated in Fig.1(b) , in order to prevent trains from out-of-range communication during operation, we further develop a cooperative consistency control method with bi-direction communication between adjacent trains. Every train except the N-th train in TP communicates with the successor in addition to the predecessor so as to obtains operation states information. In other words, every train except the N-th train is attached with perception ability to the successor. When the distance between adjacent trains is nearly out-of-range, both of them can adjust acceleration automatically to shorten the distance rather than the successor solely, which strengthens the operation security for the most part.

We allow the first train to adjust the operation state automatically by substituting ATO for RL. In order to guarantee the prescriptive schedule and route of TP, we introduce a virtual leader as a predecessor of the first train, which serves as a follower. The virtual leader is assumed to be controlled by ATO, the operation state of which is identical to the first train with uni-direction communication.

Additionally, we equip the multi-agent DQN algorithm with multi-standard rewards. Considering that out-of-range happens at the traction stage and the cruising stage abbreviated to tracru stage solely, we design different reward functions for tracru stage and braking stage respectively based on specific feature of states. The rest part of the algorithm resembles aforementioned one. We omit the details for simplicity.

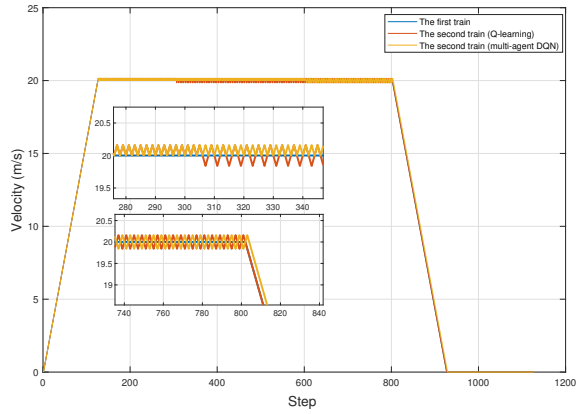
Algorithm 1 Proposed Multi-Agent DQN Algorithm for the Cooperative Consistency Control of TP

Initialize replay buffer with random numbers, Q-network, Q-target network, $\mathbf{w}_n$ with random parameters, $\mathbf{w}_n^- = \mathbf{w}_n$, $\epsilon = \epsilon_0$.
for episode e=1:E **do**
  Initialize The n-th train operation state at start, $\epsilon = \epsilon\epsilon_\mathrm{p}$ until $\epsilon = \epsilon_\mathrm{min}$.
  while $!(v_{n-1} = v_n = 0)$ **do**
    if training failure **then**
      Initialize The n-th train operation state at start.
    end if
    With probability ϵ select a random action a_n^k, otherwise select $a_n^k = arg\max_{\mathcal{A}} Q(s_n^k, a_n^k; \mathbf{w}_n)$.
    Execute action a_n^k, obtain reward r_n^k, next state s_n^{k+1}.
    Store transition $(s_n^k, a_n^k, r_n^k, s_n^{k+1})$ in replay buffer.
    step $k = k + 1$.
    if $k < I$ **then**
      Input entire sample sets to Q-network.
    else
      Every I steps select I sample sets from replay buffer.
    end if
    Perform gradient descent on (10) to update $\mathbf{w}_n$.
    Reset $\mathbf{w}_n^- = \mathbf{w}_n$ every I' steps.
  end while
end for

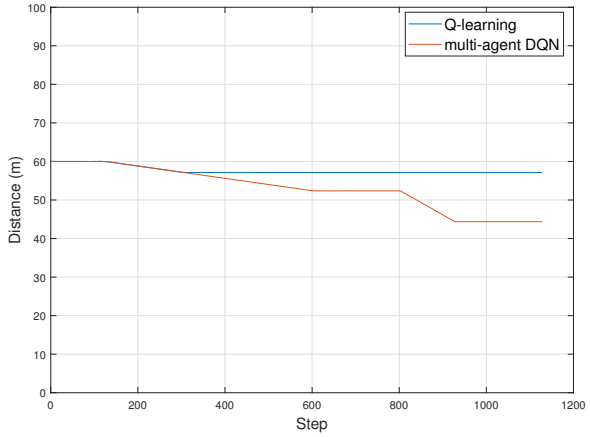
IV. Numerical Result

In this section, we consider a TP containing two trains. With the uni-direction communication, control method of the first train is based on ATO while it is based on DRL with the bi-direction communication.We refer to Q-learning as a counterpart of the proposed algorithm. The operation parameters are designed as follows: the k-th step is 0.2 s, the initial distance between two trains $x_{12}^0 = 60$ m, the maximum distance $d_\mathrm{max} = 100$ m, the goal distance $d_\mathrm{goal} = 10$ m, the initial velocity $v_1^0 = v_2^0 = 0$, the maximum velocities $v_{1,\mathrm{max}} = 20$ m/s and $v_{2,\mathrm{max}} = 20.32$ m/s. To illustrate the braking process of the second train, the cessation time of the first train lasts for 200 steps. The proposed DQN is a fully connected deep neural network consisting of a 2D input layer, a hidden layer with 100 neurons and a 3D output layer with the $\tanh(\cdot)$ activation. DQN training is set with parameter settings: initial exploration probability $\epsilon_0 = 0.1$, declining ratio $\epsilon_\mathrm{p} = 0.02$, the minimum exploration probability $\epsilon_\mathrm{min} = 0.01$, learning rate $\alpha = 0.01$, discount factor $\gamma = 0.9$, replay buffer size $B = 10,000$, sample set size $I = 50$, Q-target network replacing frequency $I' = 200$, episode quantity $E = 200$. Meanwhile, parameters

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(a)



(b)

Fig. 5: Relative distance and velocities curves of the first train and the second train. (a) Velocities curves. (b) Relative distance curves.

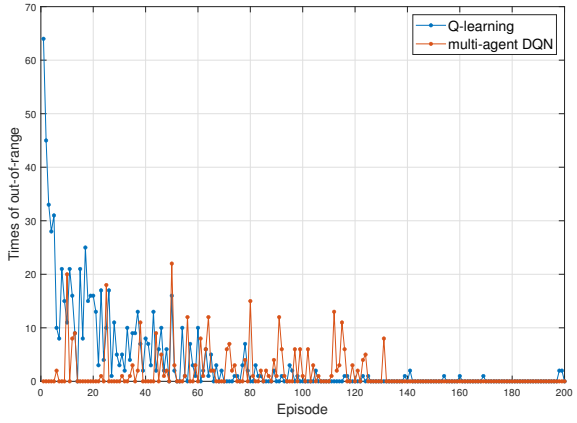


Fig. 6: Performance comparison of times of out-of-range with Q-learning and multi-agent DQN.

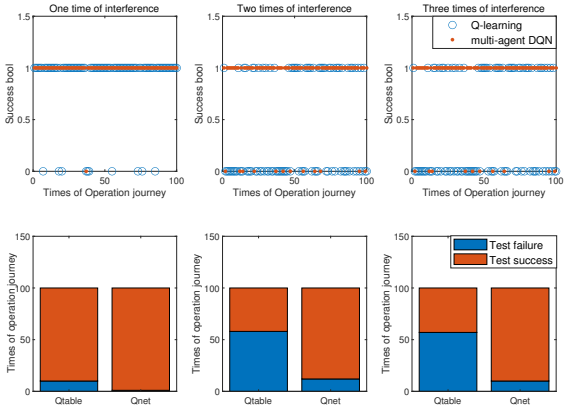


Fig. 7: Success ratio under outside interference.

of the three factors of DRL are designed. The size of distance state $X = 21$, the size of velocity state $V = 23$, the actions $a^+ = 0.8\ \mathrm{m/s^2}$ and $a^- = -0.8\ \mathrm{m/s^2}$. The reward function is set with parameter settings: fixed reward value $R = 400$, range of edge state of distance $D = 10$ m.

In Fig. 5, relative distance and velocities of the two simulated trains through the entire operation journey is illustrated. The vibration of the velocity curve of multi-agent DQN is 298 steps later than the one of Q-learning. At the braking stage, the multi-agent DQN train remains a velocity difference of 0.32 m/s. It can be observed that the second train controlled by multi-agent DQN is closer to the operation goal. This is because the multi-agent DQN train learns to sacrifice less velocity reward value to attain more distance reward value, obtaining shorter relative distance ultimately compared with the Q-learning train.

In Fig. 6, the influence on the out-of-range reduction based on multi-agent DQN versus Q-learning is presented. It is notable that multi-agent DQN effectively reduces out-of-range accidents at the beginning of the training. Meanwhile, in every successful episode, multi-agent DQN reduces out-of-range accidents by 57% evenly compared with Q-learning. This is because multi-agent DQN has greater efficiency in training, so that it learns to avoid large penalty and obtain sufficient reward in high efficiency.

In Fig. 7, the capability of anti-interference based on multi-agent DQN versus Q-learning is shown. After 200 episodes of training, the Q-network and Q-table are employed for 100 journeys, during every of which interference is added as tests. Every time of interference lasts for 100 steps. With times of interference increasing, the number of successful arrivals is declining. Under the same interference, the number of successful arrivals based on multi-agent DQN is larger than that of Q-learning to a great extent. Under three times of interference, success ratio of multi-agent DQN is 90% compared with 43%

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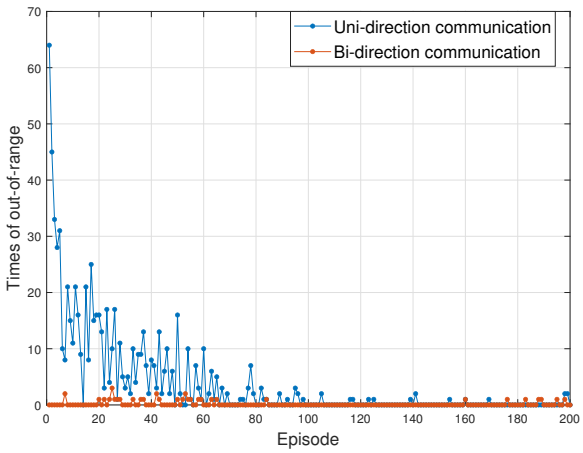


Fig. 8: Performance comparison of times of out-of-range with uni-direction and bi-direction communication.

of Q-learning. This is because multi-agent DQN is equipped with the great fitting non-linear function capability, and thus the control method derived from multi-agent DQN has greater competence in preventing outside interference.

In Fig. 8, the impact of bi-direction communication on precaution of out-of-range is demonstrated. It can be noticed that bi-direction communication method reduces the out-of-range accidents by 94.9% compared with uni-direction one. This advantage results from the perception ability of the first train. The first train obtains real-time states of the successor at every step, in which way the first train can foresee the happening of out-of-range and take precaution by deceleration in time, and constrain the relative distance effectively.

V. Conclusion

In this paper, we investigated the cooperative consistency control problem in TP system. Considering local optimization of traditional control method and a lack of operation security, a novel multi-agent DQN algorithm based on APF was proposed to solve zero prior knowledge problem and obtain the global optimal control strategy. Furthermore, the bi-direction communication method of trains was proposed to prevent TP from the out-of-range accident specifically. Simulation results showed that the proposed multi-agent DQN algorithm could not only contribute to the achievement of the operation goal, but also improve the anti-interference capability of the control method. Meanwhile, out-of-range is reduced based on the proposed algorithm. Moreover, the proposed bi-direction communication method could take precaution against the out-of-range accident for the most part.

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